Scalar subspaces and flag transitivity

Human Compression project

Abstract

The previously acquired scalar-subspace family satisfies the literal algebraic local two-transitivity condition in arbitrary rank. Combining its intersection formula with the proved descent theorem shows that these pregeometries are thick, residually connected, flag-transitive geometries. The additional argument is a surjectivity calculation on independent scalar coordinates.

Let $e \geq 3$ be odd, put $t = 2^e$ and $K = \mathbb{F}_t$, and choose $k \geq 2$ distinct odd primes $a_1, \ldots, a_k$. Set

$$A = \prod_i a_i, \quad F = \mathbb{F}_{t^A}, \quad F_i = \mathbb{F}_{t^{A/a_i}}, \quad G = \mathrm{Sz}(t^A), \quad S_i = \mathrm{Sz}(F_i).$$

Choose $x_i \in \mathbb{F}_{t^{a_i}} \setminus K$ and put

$$T = \mathrm{span}_K(1, x_1, \ldots, x_k), \qquad Q = Z_T B_t,$$

where $B_t = U_t H_t$ is the standard point stabilizer of $\mathrm{Sz}(t)$ and Z_T is the central unipotent subgroup with parameter in T. The earlier scalar-subspace proof establishes that $1, x_1, \ldots, x_k$ are independent over K and, for $J \subseteq \{1, \ldots, k\}$,

$$T \cap F_J = T_J = \mathrm{span}_K(1, x_i : i \notin J), \qquad Q \cap \bigcap_{j \in J} S_j = Z_{T_J} B_t, \tag{1}$$

where $F_J = \mathbb{F}_{t^{A/\prod_{j \in J} a_j}}$. For completeness, apply $\sigma_i : y \mapsto y^{t^{A/a_i}}$ to a linear combination lying in F_i. This automorphism fixes every x_j with $j \neq i$, fixes K, and moves x_i. Subtracting the two expressions forces its x_i coefficient to vanish. The same argument with a zero linear combination proves independence, and intersection over $i \in J$ gives the displayed formula. The coordinate multiplication law makes $Z_T B_t$ a subgroup: U_t centralizes Z_T, and H_t acts by all nonzero K-scalars.

Theorem 1. *The coset system $\Gamma(G; S_1, \ldots, S_k, Q)$ is a thick, residually connected, flag-transitive geometry of rank $k + 1$. It satisfies the source's residual weak primitivity condition and local two-transitivity, now on its actual rank-one residues.*

Proof. Put $T_i = \mathrm{span}_K(1, x_j : j \neq i)$ and $H_i = Q \cap S_i = Z_{T_i} B_t$. Every left H_i-coset in Q has a representative in Z_T, and

$$T/T_i \longrightarrow Q/H_i, \qquad b + T_i \longmapsto z(b) H_i \tag{2}$$

is a bijection of sets. Indeed $Q = Z_T H_i$ and $Z_T \cap H_i = Z_{T_i}$, by (1). No quotient-group structure on Q/H_i is needed. Given arbitrary coordinates $c_i \in K$, the single vector $\sum_i c_i x_i \in T$ represents them in every T/T_i. Thus

$$Q \longrightarrow \prod_i Q/H_i$$

is surjective.

The standard subfield cosets have the common-representative property by the acquired descent theorem, with Steinberg endomorphism ρ^e and independent relative degrees a_i. Its attachment lemma and (2) therefore give that property for all the types including Q. Every flag extends to a translate of the standard chamber; hence this is a flag-transitive geometry. In particular all its actual residues are isomorphic to the standard subgroup-intersection residues.

The earlier proof of residual weak primitivity can now be read in these actual residues. Before selecting Q, an unused field type makes a maximal prime-degree inclusion of simple Suzuki groups, since the bottom field exponent is at least $e > 1$. After every field type has been selected, the remaining inclusion is $B_t < \mathrm{Sz}(t)$, which is maximal. Once Q is selected, an unused field type replaces T_J by a hyperplane containing K. The overgroups of $Z_{T_J \cap F_i} B_t$ inside $Z_{T_J} B_t$ correspond to K-subspaces between those two parameter spaces: the proof is the elementary equality $L = (L \cap Z_{T_J}) B_t$, together with the scalar action of H_t. The codimension is one, so every such inclusion is maximal.

The full intersection is B_t. The panel missing Q has $[\mathrm{Sz}(t) : B_t] = t^2 + 1$ chambers and the usual doubly transitive Suzuki action. The panel missing S_i has $[Z_{K+Kx_i} B_t : B_t] = t$ chambers. Its quotient action on $(K + Kx_i)/K$ contains all translations and all nonzero K-scalars; it is the doubly transitive affine action on K. Since $t \geq 8$, every panel has at least three chambers. This proves thickness and actual local two-transitivity.

Finally consider a standard residue of rank at least two. If Q has been selected, at least two field types remain. Their two parameter hyperplanes sum to T_J, so their subgroups generate $Z_{T_J} B_t$. If Q is unselected and at least two field types remain, two distinct maximal simple subfield groups generate the current Suzuki group. If precisely one field type and Q remain, the base $\mathrm{Sz}(t)$ is maximal in $\mathrm{Sz}(t^{a_i})$, and $Z_{K+Kx_i} B_t$ is not contained in it because $x_i \notin K$. Again they generate the residue group. The coset incidence graph is connected exactly when its parabolics generate; flag transitivity transports this conclusion to every residue. Thus the geometry is residually connected. □

Attribution and source scope. The scalar family and its intersection proof were already independently accepted in the project; the Frobenius-isolation step there was suggested by the root reviewer. The descent and coset-attachment lemmas are earlier project proofs retaining classical Lang–Steinberg. The new step here is the explicit coordinate bijection (2) and its geometric consequences. It uses no new finite acquisition and does not change any frozen proof.

The fixed higher-rank source, D. Leemans, Note di Matematica 20(1) (2000/2001), p. 2, quotes the earlier assertion that no rank at least four geometry satisfies both residual weak primitivity and local two-transitivity. Taking $k \geq 3$ disproves this statement even with firmness, residual connectedness and flag transitivity imposed. For $k = 2$, the rank is three while the ambient field degree is composite, and every panel is thick. These are further consequences of the same already acquired family, not new original or principal-refutation counts. No full Lean formalization or complete classification is claimed.